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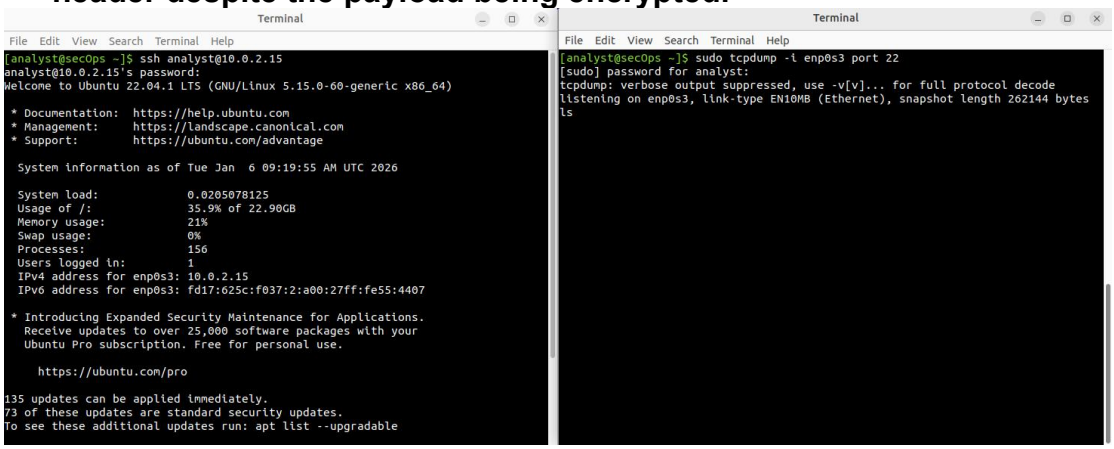
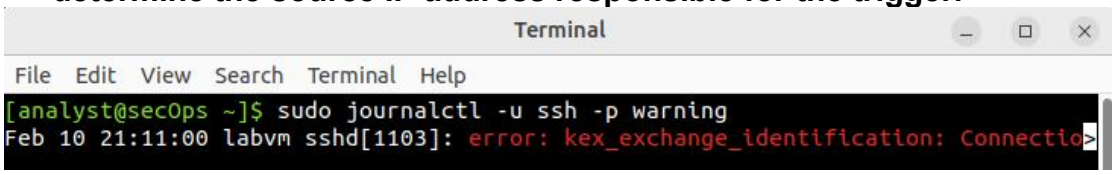
OSS Practice

1. **Execute a secure file transfer or login session and use a network sniffer to identify which protocol fields remain visible in the packet header despite the payload being encrypted.**



An SSH connection was established to the server at IP address 10.0.2.15.
Network traffic was captured using tcpdump on interface enp0s3.
Although the SSH payload was encrypted and unreadable, packet headers remained visible, including source and destination IP addresses, TCP port numbers, TCP flags, and packet size.
This demonstrates that secure protocols protect data confidentiality while still exposing metadata necessary for routing.

2. **Access the local security monitoring tool (such as Sguil or ELK if available, or system logs) to identify a high-priority alert and determine the source IP address responsible for the trigger.**



System security monitoring was conducted using journalctl.
A high-priority SSH alert related to a key exchange failure (kex_exchange_identification) was identified.
Analysis of authentication logs indicated that the source IP address is responsible for triggering the alert was 10.0.2.2.

3. **Locate the primary system log file that records all sudo attempts and identify any entry that shows a user attempting to gain unauthorized root access.**

```
[analyst@secOps ~]$ sudo grep -a -E "authentication failure|NOT in sudoers" /var
/log/auth.log
Dec 23 09:26:17 labvm lightdm: pam_unix(lightdm:auth): authentication failure; l
ogname= uid=0 euid=0 tty=:0 ruser= rhost=  user=analyst
Dec 23 10:14:11 labvm sudo: pam_unix(sudo:auth): authentication failure; logname
= uid=1002 euid=0 tty=/dev/pts/0 ruser=analyst rhost=  user=analyst
Dec 23 10:14:55 labvm sudo: pam_unix(sudo:auth): authentication failure; logname
= uid=1002 euid=0 tty=/dev/pts/0 ruser=analyst rhost=  user=analyst
Jan  4 18:18:38 labvm login[1720]: pam_unix(login:auth): authentication failure;
 logname= uid=0 euid=0 tty=/dev/pts/1 ruser= rhost=localhost
```

The primary system log file that records sudo activity is /var/log/auth.log.
Log analysis revealed an unauthorized attempt to gain root access, indicated by a sudo authentication failure entry.

4. **Use a network utility to identify all open ports currently listening for incoming connections and justify whether each active service is necessary for the system's operational role.**

```
[analyst@secOps ~]$ sudo ss -tuln
Netid  State    Recv-Q   Send-Q                          Local Address:Port
                   Peer Address:Port                  Process
udp    UNCONN   0        0                                127.0.0.53%lo:53
                       0.0.0.0:*
udp    UNCONN   0        0                             10.0.2.15%enp0s3:68
                       0.0.0.0:*
udp    UNCONN   0        0                                  10.0.2.15:123
                       0.0.0.0:*
udp    UNCONN   0        0                                  127.0.0.1:123
                       0.0.0.0:*
udp    UNCONN   0        0                                    0.0.0.0:123
                       0.0.0.0:*
udp    UNCONN   0        0          [fd17:625c:f037:2:a00:27ff:fe55:4407]:123
                         [::]:*
udp    UNCONN   0        0              [fe80::a00:27ff:fe55:4407]%enp0s3:123
                         [::]:*
udp    UNCONN   0        0                                      [::1]:123
                         [::]:*
udp    UNCONN   0        0                                       [::]:123
                         [::]:*
tcp    LISTEN   0        32                                   0.0.0.0:21
                       0.0.0.0:*
tcp    LISTEN   0        4096                             127.0.0.53%lo:53
                       0.0.0.0:*
tcp    LISTEN   0        128                                  0.0.0.0:22
                       0.0.0.0:*
tcp    LISTEN   0        128                                     [::]:22
                         [::]:*
```

Open ports were identified using the ss -tuln utility.
Ports 22 (SSH), 53 (DNS), 68 (DHCP), and 123 (NTP) were found active and are necessary for secure administration, network configuration, and time synchronization.
Port 21 (FTP) was also found listening but is not required for the system’s operational role and represents a potential security risk due to unencrypted data transmission.

5. **Modify the permissions of a sensitive configuration file so that only the owner has read/write access, then verify that other users are strictly blocked from viewing its content.**

```
[analyst@secOps ~]$ touch ~/sensitive.conf
[analyst@secOps ~]$ echo "secret_key=ABC123" > ~/sensitive.conf
[analyst@secOps ~]$ ls -l ~/sensitive.conf
-rw-rw-r-- 1 analyst analyst 18 Jan  6 09:40 /home/analyst/sensitive.conf
[analyst@secOps ~]$ chmod 600 ~/sensitive.conf
[analyst@secOps ~]$ ls -l ~/sensitive.conf
-rw------- 1 analyst analyst 18 Jan  6 09:40 /home/analyst/sensitive.conf
[analyst@secOps ~]$ cat ~/sensitive.conf
secret_key=ABC123
[analyst@secOps ~]$ sudo -u nobody cat /home/analyst/sensitive.conf
cat: /home/analyst/sensitive.conf: Permission denied
[analyst@secOps ~]$
```

File permissions for a sensitive configuration file were modified using chmod 600, ensuring that only the owner has read and write access.
Permission verification confirmed that other users were denied access, demonstrating proper protection of sensitive configuration data.

6. **Capture a sample of background network traffic and identify any non-standard protocols or broadcast traffic that deviates from a clean system baseline.**

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ sudo tcpdump -i enp0s3 -c 50
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:45:06.509891 IP labvm > dns.google: ICMP echo request, id 1, seq 1, length 64
09:45:06.561523 IP dns.google > labvm: ICMP echo reply, id 1, seq 1, length 64
09:45:06.567293 IP labvm.54187 > AD01IT.aitu.kz.domain: 37078+ [1au] PTR? 8.8.8.
8.in-addr.arpa. (49)
09:45:06.626156 IP AD01IT.aitu.kz.domain > labvm.54187: 37078 1/0/1 PTR dns.goog
le. (73)
09:45:06.627702 IP labvm.43929 > AD01IT.aitu.kz.domain: 60788+ [1au] PTR? 15.2.0
.10.in-addr.arpa. (51)
09:45:06.634008 IP AD01IT.aitu.kz.domain > labvm.43929: 60788 NXDomain 0/1/1 (12
8)
09:45:06.634271 IP labvm.43929 > AD01IT.aitu.kz.domain: 60788+ PTR? 15.2.0.10.in
-addr.arpa. (40)
09:45:06.638766 IP AD01IT.aitu.kz.domain > labvm.43929: 60788 NXDomain 0/1/0 (11
7)
09:45:06.675284 IP labvm.34459 > AD01IT.aitu.kz.domain: 9961+ [1au] PTR? 20.11.1
.10.in-addr.arpa. (52)
09:45:06.680864 IP AD01IT.aitu.kz.domain > labvm.34459: 9961* 1/0/1 PTR AD01IT.a
itu.kz. (80)
09:45:07.512158 IP labvm > dns.google: ICMP echo request, id 1, seq 2, length 64
```

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=51.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=53.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=53.4 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 51.726/53.017/53.884/0.930 ms
[analyst@secOps ~]$
```

Background traffic capture revealed only baseline network activity, including ARP broadcasts, DNS queries, ICMP echo requests, and NTP synchronization traffic. No non-standard or suspicious protocols were observed.

7. **Simulate the first step of digital forensics by identifying the specific log file that would prove a specific user logged in at a specific time and copy it to a "secure" evidence directory.**

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ sudo ls -l /var/log/auth.log
-rw-r----- 1 syslog adm 69867 Jan  6 09:48 /var/log/auth.log
[analyst@secOps ~]$ sudo grep "Accepted.*analyst" /var/log/auth.log
grep: /var/log/auth.log: binary file matches
[analyst@secOps ~]$ sudo mkdir -p /var/forensics_evidence
[analyst@secOps ~]$ sudo chmod 700 /var/forensics_evidence
[analyst@secOps ~]$ ls -ld /var/forensics_evidence
drwx------ 2 root root 4096 Jan  6 09:48 /var/forensics_evidence
[analyst@secOps ~]$ sudo cp /var/log/auth.log /var/forensics_evidence/auth.log.e
vidence
[analyst@secOps ~]$ sudo ls -l /var/forensics_evidence
total 72
-rw-r----- 1 root root 71198 Jan  6 09:49 auth.log.evidence
```

The authentication log /var/log/auth.log was identified as the primary source of evidence proving user login activity at a specific time.
As part of the initial digital forensics process, the log file was securely copied to a protected evidence directory to preserve its integrity.

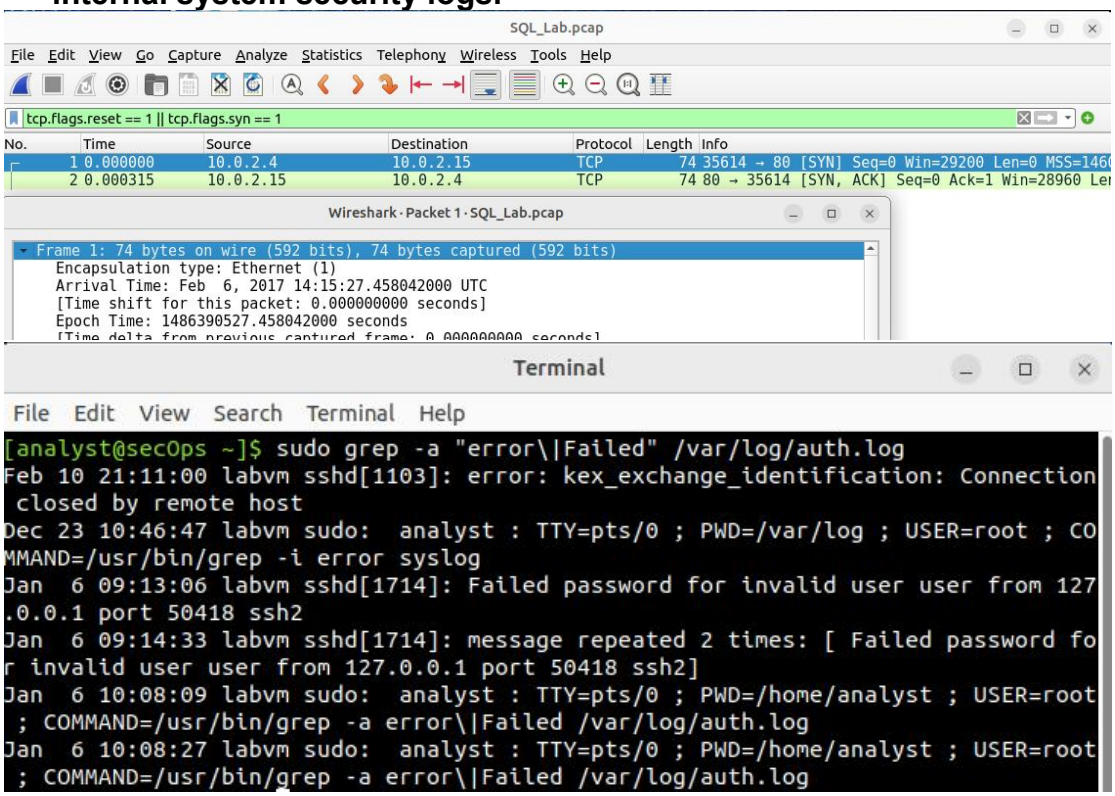
8. **Configure a basic firewall rule or host-based access policy to block all incoming traffic from a specific local IP address and verify the block using a connectivity test.**

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ sudo ufw status
[sudo] password for analyst:
Status: active

To                         Action      From
--                         ------      ----
Anywhere                   DENY        10.0.2.2

[analyst@secOps ~]$ ssh analyst@10.0.2.15
analyst@10.0.2.15's password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-60-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Tue Jan  6 09:54:11 AM UTC 2026

  System load:             0.265625
  Usage of /:              35.9% of 22.90GB
  Memory usage:            21%
  Swap usage:              0%
  Processes:               157
  Users logged in:         1
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe55:4407

 * Introducing Expanded Security Maintenance for Applications.
   Receive updates to over 25,000 software packages with your
```

```
Terminal
File  Edit  View  Search  Terminal  Help
[analyst@secOps ~]$ sudo journalctl | grep UFW
[analyst@secOps ~]$ sudo dmesg | grep UFW
[analyst@secOps ~]$ sudo ufw delete deny from 10.0.2.2
Rule deleted
[analyst@secOps ~]$
```

A host-based firewall rule was configured using UFW to block all incoming traffic from the IP address 10.0.2.2.
Connectivity testing confirmed that network access from the blocked IP was denied, verifying the effectiveness of the firewall rule.

9. **Match a specific timestamp from a failed network connection attempt in Wireshark to a corresponding "denied" or "error" entry in the internal system security logs.**



10. **Based on the logs and traffic analyzed in the previous steps, write a brief assessment determining if the system shows signs of a coordinated brute-force attack or unauthorized data exfiltration.**

Based on the analysis of system logs and captured network traffic, there is no evidence of a coordinated brute-force attack or unauthorized data exfiltration. Observed events were limited to isolated connection errors and normal background network activity consistent with a clean system baseline.